

Topic: Apply bucket policies and enable server side encryption

1. Create a new storage bucket in your chosen cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) named 'my-music-uploads' and upload any 2 MP3 files to it.

Ans-

Step 1.1 — Open S3

1. Login to AWS Management Console.
2. Search for S3.
3. Open S3 – Scalable Storage.
4. Click Create bucket.

Step 1.2 — Enter bucket name

Step 1.3 — Select region

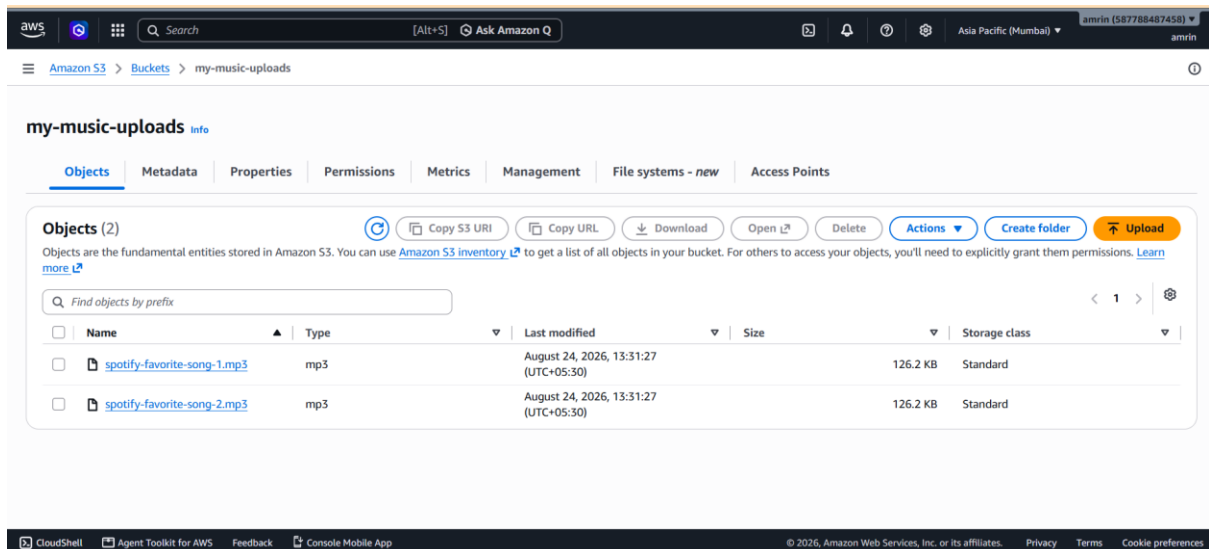
Step 1.4 — Object Ownership

Step 1.5 — Block Public Access

Step 1.6 — Bucket Versioning

Step 1.7 — Create bucket

Step 1.8 — Upload MP3 files



**2. Apply a bucket policy to 'my-music-uploads' so only users with a specific email address (your own or a friend's) can download files from the bucket.

Hint: Use the bucket's access management settings to add an IAM user or assign a permission for a specific email.**

Ans-

Step 1 — Create IAM User

1. Search for IAM.
2. Open IAM.
3. Click Users.
4. Click Create user.

Step 2 — Give the user S3 permissions

IAM → Policies → Create policy

Choose the JSON editor.

Use:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowMusicBucketList",
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::my-music-uploads"
    },
    {
      "Sid": "AllowMusicDownload",
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-music-uploads/*"
    }
  ]
}
```

Step 2.3 — Attach policy

Go to: IAM → Users → music-download-user

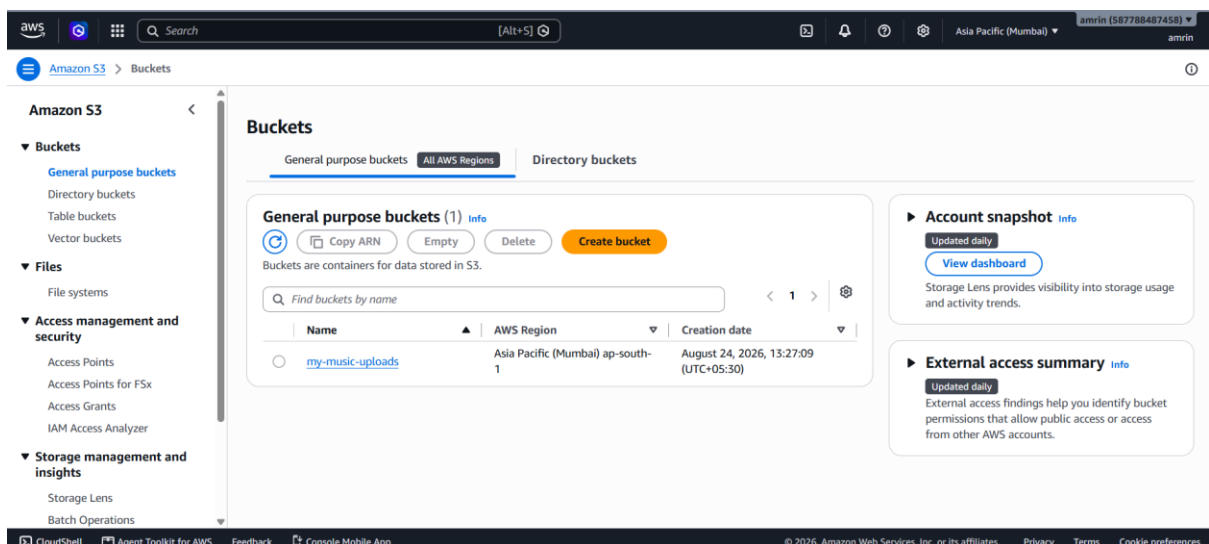
Then: Permissions → Add permissions → Add permissions

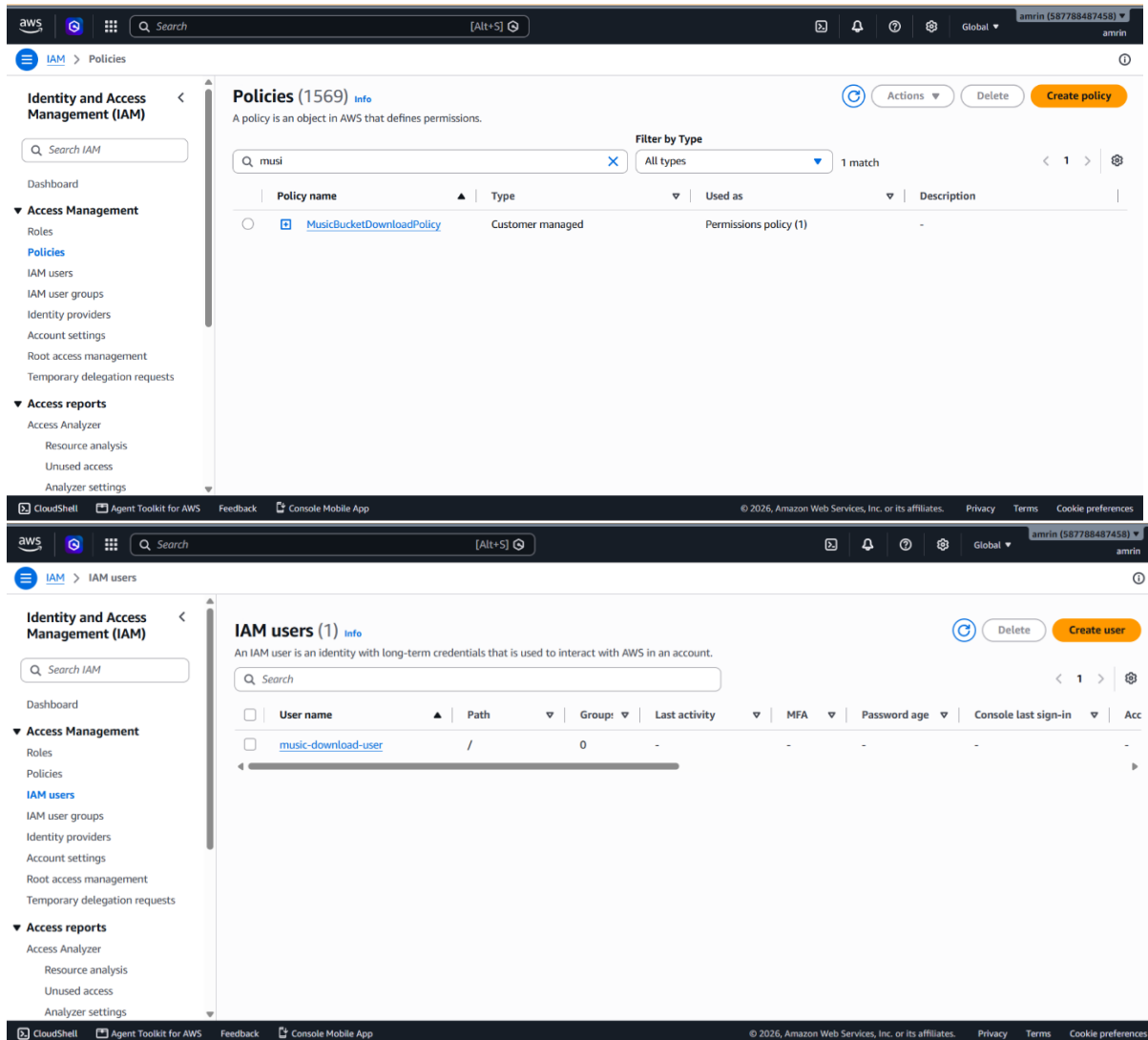
Attach: MusicBucketDownloadPolicy

Step 2.4 — Create bucket policy: S3 → my-music-uploads → Permissions

Use this policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowSpecificUserDownload",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::YOUR_ACCOUNT_ID:user/music-download-user"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-music-uploads/*"
    }
  ]
}
```





3. Enable server-side encryption for all objects uploaded to 'my-music-uploads', ensuring that future uploads are encrypted automatically by the cloud provider.

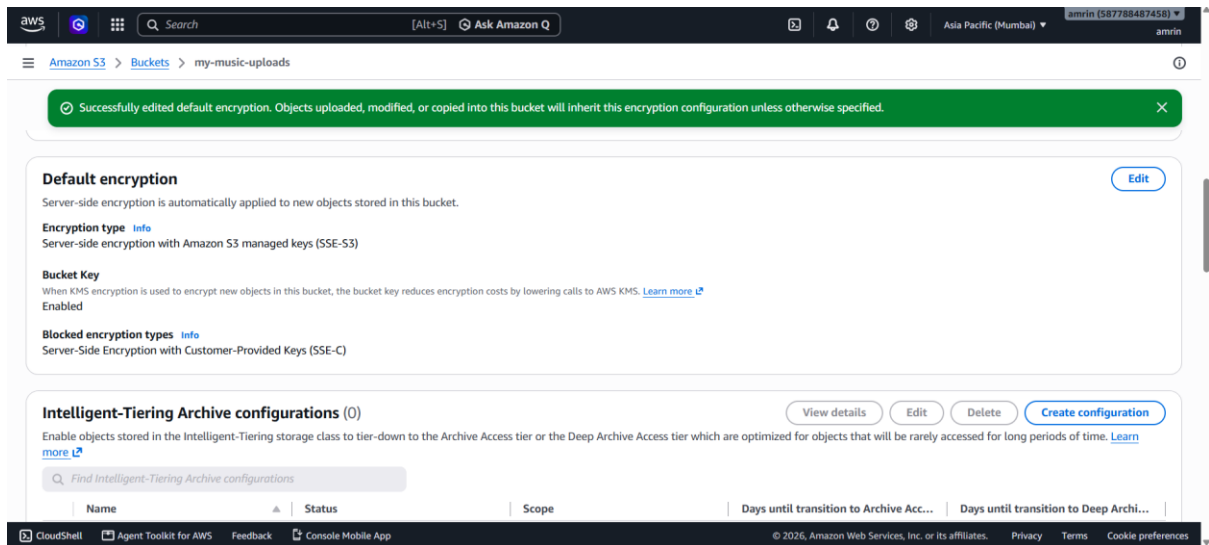
Ans- Step 3.1 — Open bucket properties

Go to: S3 → my-music-uploads → Properties

Step 3.2 — Select encryption

Choose: Server-side encryption with Amazon S3 managed keys (SSE-S3)

Step 3.3 — Verify



4. Test your bucket's security by trying to access an uploaded MP3 file from a browser where you are not logged in — take a screenshot of the error or access denied message.

Ans-

Step 4.1 — Get the MP3 URL

Go to:

S3 → my-music-uploads → Objects

Open:

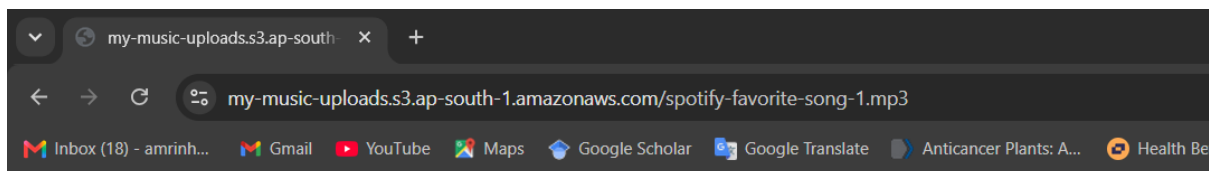
spotify-favorite-song-1.mp3

Step 4.2 — Open private/incognito browser

On Chrome:

Ctrl + Shift + N

Open an Incognito window.



This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>V6Q5JD8QW96KXZQ</RequestId>
  <HostId>k6HJ6gt6n/sDzRmpQuj70Dh8i/+KRAz8Ju+IKIcH7p/FAWYyqXXn7CF0G2NwBsQEuvUlcQ2wThvicqy555dImo0bFdZg7idhK</HostId>
</Error>
```

5. Use an AI tool like ChatGPT or Copilot to generate a sample bucket policy that restricts access to your bucket by IP address, then modify and apply it to 'my-music-uploads'.
Hint: Copy the AI-generated policy, review it for your cloud provider's syntax, and adjust the IP to your college's or your own network.

Ans-

Step 1 — Find your public IP

Step 2 — Open your S3 bucket

Click Buckets.

Open: my-music-uploads

Click Permissions.

Step 3 — Open Bucket Policy

Scroll to Bucket policy.

Click Edit.

You will see the JSON policy editor.

Step 4 — Generate the AI policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyRequestsOutsideAllowedIP",
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-music-uploads/*",
      "Condition": {
        "NotIpAddress": {
          "aws:SourceIp": "YOUR_PUBLIC_IP/32"
        }
      }
    }
  ]
}
```

}

Step 5 — Replace the IP address

Step 6 — Apply the policy

Step 7 — Test from your allowed network

Step 8 — Test from another IP

The screenshot shows the Amazon S3 console interface for editing a bucket policy. The left sidebar contains navigation links for Buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, IAM Access Analyzer for S3, Storage Lens, Dashboards, Storage Lens groups, AWS Organizations settings, and Feature spotlight. The main content area is titled 'Edit bucket policy' for the bucket 'my-music-uploads'. It includes a 'Bucket policy' section with a JSON editor showing the following policy:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "AllowSpecificUserDownload",
6       "Effect": "Allow",
7       "Principal": {
8         "AWS": "arn:aws:iam::123456789012:user/music-download-user"
9       },
10      "Action": "s3:GetObject",
11      "Resource": "arn:aws:s3::my-music-uploads/*"
12    },
13    {
14      "Sid": "DenyRequestsOutsideAllowedIP",
15      "Effect": "Deny",
16      "Principal": "*",
17      "Action": "s3:GetObject",
18      "Resource": "arn:aws:s3::my-music-uploads/*",
19      "Condition": {
20        "NotIpAddress": { "aws:SourceIp": "203.0.113.50/32" }
21      }
22    }
23  ]
24 }
```

Below the JSON editor is an 'Edit statement' panel with a 'Select a statement' section and an 'Add new statement' button. A green notification bar at the bottom of the console indicates 'Bucket policy successfully updated.'